Antony Hubert

Mechanical Engineer (M.Sc.) | Development, Process Engineering, Design and Simulation

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Work permit: German Job Seeker Visa, eligible for EU Blue Card. Available immediately.

PROFILE

Mechanical engineer with an M.Sc. in Advanced Materials and Manufacturing and a peer-reviewed publication in Procedia CIRP. I develop and qualify manufacturing processes from first trials through to a documented, repeatable production window, combining Design of Experiments, CAD, FEA and CFD with laboratory analysis I carry out myself. Experience spans metal additive manufacturing (PBF-LB/M), design in CATIA V5, Siemens NX and PTC Creo, simulation in ANSYS, and 4-axis CNC production. Based in 73430 Aalen and open to relocation.

CORE COMPETENCIES

Development and Research: Design of Experiments (DoE), Taguchi method, ANOVA, regression analysis, parameter studies, process qualification, prototyping, technical documentation, peer-reviewed publication

Design and CAD: CATIA V5, Siemens NX, PTC Creo, AutoCAD, GD&T (ISO 1101), reverse engineering, automotive body-in-white, EV component layout, concept sketching, CAD rendering

Simulation (CAE): ANSYS Fluent (CFD), ANSYS Mechanical and APDL (FEA), flow analysis, thermal analysis

Manufacturing: 4-axis CNC machining and CAM, toolpath optimisation, additive manufacturing (PBF-LB/M, L-PBF, SLM, DMLS), Design for Additive Manufacturing (DfAM), build preparation (Materialise Magics, Autodesk Netfabb)

Materials and Testing: Metallography, optical and scanning electron microscopy (SEM), atomic force microscopy (AFM), surface metrology (KEYENCE VR-3100, MarSurf M400), dilatometry and TMA (NETZSCH DIL 402, METTLER TOLEDO), heat treatment, porosity and grain analysis, micro-CT. Materials: FeSi6.5, AlSi10Mg, cobalt-chromium

Ways of Working: Project coordination, training and mentoring, cross-team collaboration, technical reporting, Python (foundation), machine learning fundamentals

PROFESSIONAL EXPERIENCE

Academic Research Assistant, Process Development **09/2023 – 03/2024**
Hochschule Aalen (Aalen University), LaserApplication Center, Aalen, Germany

- Led development of a laser-based manufacturing process for a brittle magnetic alloy, from first trials to a documented, repeatable production window. Results published in Procedia CIRP (2024).
- Reduced surface roughness by approximately 60 percent at 0.152 percent porosity using an in-process laser remelting method developed in this role.
- Performed all laboratory analysis in-house: metallography, optical microscopy, SEM and surface metrology. Coordinated with the team to run experiments and deliver on schedule.
- Completed design and commissioning of a shielding-gas circulation system with heated build platform, begun in the preceding working student role.

Working Student, Manufacturing Engineering **05/2023 – 08/2023**
Hochschule Aalen (Aalen University), LaserApplication Center, Aalen, Germany

- Designed a shielding-gas circulation system in CATIA V5 and validated the design by CFD analysis in ANSYS Fluent.
- Operated metal additive manufacturing systems end to end: CAD, build preparation, machine operation, post-processing and inspection.
- Reduced failed builds through improved support design and parameter selection.

Application Engineer, CAD and CAE **03/2022 – 09/2022**
CADD Centre Training Services Pvt. Ltd., Chennai, India

- Trained and mentored over 100 engineers and students in CATIA V5, Siemens NX, PTC Creo, ANSYS and AutoCAD.
- Designed automotive body-in-white structures and electric-vehicle components, and advised clients on manufacturing feasibility.

CAD and CAM Engineer, CNC Manufacturing **06/2021 – 02/2022**
Axis Automation Pvt. Ltd., Chennai, India

- Programmed and operated 4-axis CNC machining centres for precision automotive components.
- Reverse engineered legacy parts from physical samples.
- Optimised toolpath strategies to increase output and reduce material waste.

EDUCATION

M.Sc. Advanced Materials and Manufacturing **10/2022 – 09/2025**

Hochschule Aalen (Aalen University), Aalen, Germany. Final grade 2.7 (German scale, 1.0 best)

- Thesis: optimisation of layer surface topography in FeSi6.5 soft magnetic components through in-process laser remelting in PBF-LB/M. Supervisors: Prof. Dr. Harald Riegel, Prof. Dr. Dagmar Goll.

B.E. Mechanical Engineering **07/2017 – 08/2021**

Jeppiaar Institute of Technology, Chennai, India. CGPA 8.5 / 10 (Indian scale, 10 best)

- Thesis: porosity reduction in additively manufactured cobalt-chromium for medical implants, using a Taguchi L9 design of experiments.

PROJECTS

- Thermal expansion of additively manufactured AlSi10Mg (2023/24, Hochschule Aalen). Designed a 30-specimen geometry matrix, ran heat treatment at 300 °C, and measured expansion to 400 °C by dilatometry and TMA. Supervisor: Prof. Dr. Markus Merkel.
- Shielding-gas circulation system for a PBF-LB/M machine. CATIA V5 design, ANSYS Fluent CFD analysis, manufacture, integration and commissioning.
- Camera mounting bracket for a laser processing head. Designed and additively manufactured in-house to give repeatable positioning for in-process monitoring.
- Sustainable product development with additive manufacturing. Life-cycle comparison of additive and conventional manufacturing.
- Concept vehicle exterior design, Automotive Styling Boot Camp 2019, Bangalore. Winning team.

PUBLICATION

Hofele M., Antony H., Kolb D., Riegel H. (2024). Production of soft magnetic FeSi6.5 parts with precise layer topography using integrated laser remelting at PBF-LB/M to induce multi-material lamination. Procedia CIRP, Volume 124, pages 6 to 11. DOI: 10.1016/j.procir.2024.08.060

LANGUAGES

- English: C1, full professional proficiency. Language of my Master's study, thesis and publication.
- German: B1. B2 examination scheduled for December 2026.
- Tamil: Native speaker.

CERTIFICATIONS AND FURTHER TRAINING

- CAE and CFD using ANSYS. CIPET, Chennai, 2019
- CAD/CAM using PTC Creo. CIPET, Chennai, 2019
- CAD/CAM using CATIA V5. CIPET, Chennai, 2019
- Applied Ergonomics. NPTEL / IIT Madras, 2020
- Automotive Styling Boot Camp, 80 hours. ExpertsHub, Bangalore, 2019
- Independent study in machine learning and Python for manufacturing, 2025 to present

VOLUNTEERING AND ENGAGEMENT

- Volunteer, German Red Cross, 30 hours. CONTRIBUTE programme, Aalen University, 2022/23.
- Student Coordinator, TECH AGRONA 2K20 national project expo, Jeppiaar Institute of Technology.